

Unit 7 — The Firm and its Customers

Costs, demand, profit maximization & market power

A companion study guide and running lecture notes. Every cost curve, isoprofit diagram, marginal-revenue graph and worked example from the slides is redrawn here so you can follow along visually while the lecturer talks. Definitions, intuition boxes, worked solutions and exam tips are added throughout. Academic year 2025–2026.

HOW TO USE THESE NOTES

Skim the **definition boxes** before each lecture. During the lecture, follow the **redrawn diagrams** — they use the same axes and colours as the slides. Colour convention used throughout: **red = demand / marginal revenue**, **magenta = marginal cost**, **teal = average cost**. After the lecture, test yourself with the **self-test questions** at the end.

COVERAGE NOTE — PLEASE READ

These notes are built from the two uploaded slide decks: **Part 1** (slides 1–20: Introduction & Production/Costs) and **Part 2** (slides 40–51: Profit Maximization). The slides in between — **Section C “Demand, elasticity and revenue”** (≈ slides 21–39) and the detailed slides for **E “Gains from Trade”** and **F “Competition, differentiation and market power”** — were not in the uploaded files. Because Part 2 *relies* on those ideas (elasticity, the demand curve, marginal revenue), the sections marked “*bridging*” below reconstruct them in the standard CORE/KU Leuven treatment so your notes stay coherent. Cross-check them against the missing slides when you get them.

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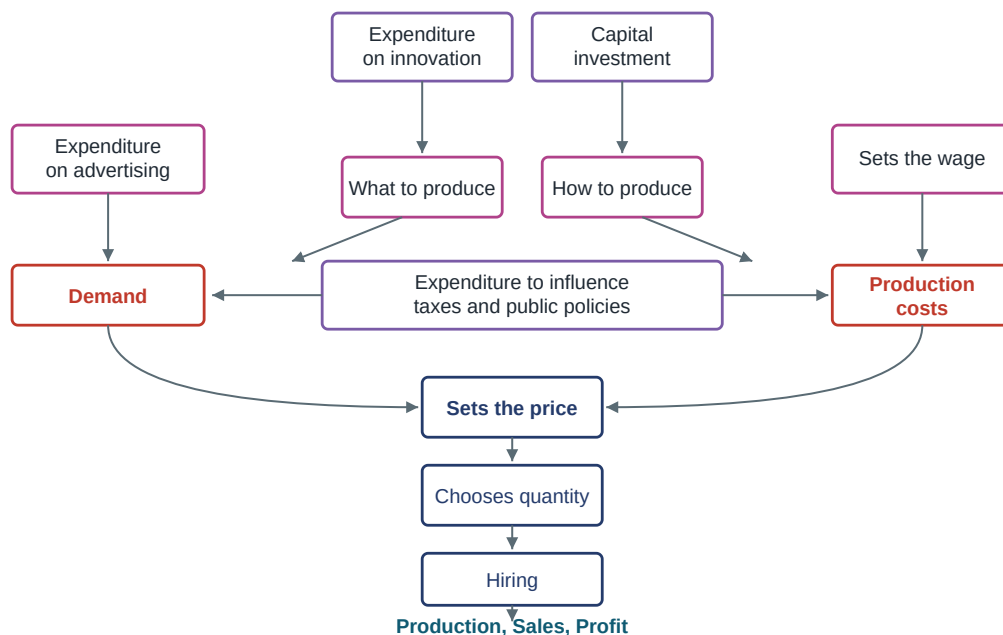
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A. Introduction — the firm and its decisions

A firm is a web of decisions. This unit zooms in on two of them — **what price to set** and **how much to produce** — and shows they depend on **demand** (how customers respond to price) and on **production costs**.

The firm's decisions

The firm's many choices feed into each other and ultimately into production, sales and profit:



Two inputs are decisive for pricing and output: **Demand** (left) and **Production costs** (right). Everything else — innovation, advertising, wages, lobbying — ultimately works through those two channels to “Sets the price → Chooses quantity → Hiring”.

What this unit is about

- Key firm decisions: **what price to set** and **what quantity to produce**.
- These depend on the **demand function** (how responsive quantity sold is to price — its **price elasticity**) and the **cost function** (how costs change with output).
- We model firms that produce **differentiated products** and therefore have some **market power** (they face a downward-sloping demand curve, unlike price-takers).
- We measure the **gains from trade** (surplus) and ask how **policy** can affect its division between firms and customers.

THE CORE TRADE-OFF IN ONE LINE

Setting a **lower** price raises quantity sold but lowers revenue per unit; a **higher** price does the reverse. Whether a price change raises or lowers *revenue* depends on the **elasticity of demand**. Whether it raises *profit* also depends on the **cost of producing** the extra unit (marginal cost). Profit maximization is about balancing these.

B. Production & costs — key concepts

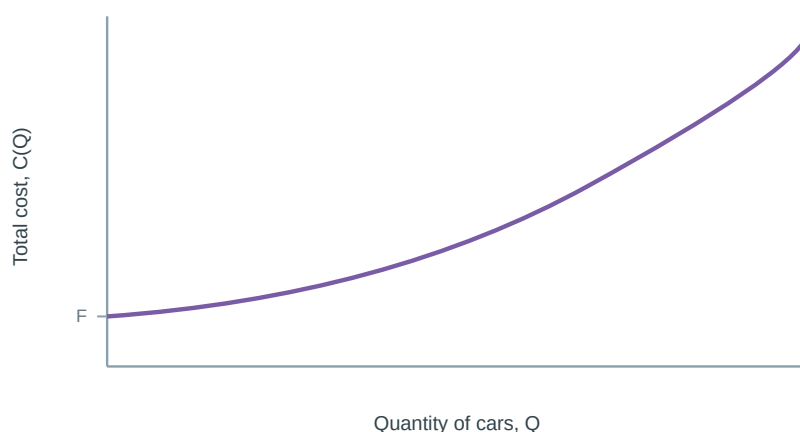
To choose price and output well, managers must know how **costs vary with the quantity produced**. That relationship is the **cost function**.

Fixed, variable and total cost

DEFINITIONS

- **Cost function $C(Q)$** — shows how *total* production cost varies with quantity Q produced.
- **Fixed costs** — do not vary with output (e.g. opportunity cost of capital, R&D). They can depend on the planning horizon (long run vs short run). “Fixed” means *not varying with the production rate*, not literally unchangeable.
- **Variable costs** — increase with output (e.g. wages, materials).

Total cost of production, $C(Q)$



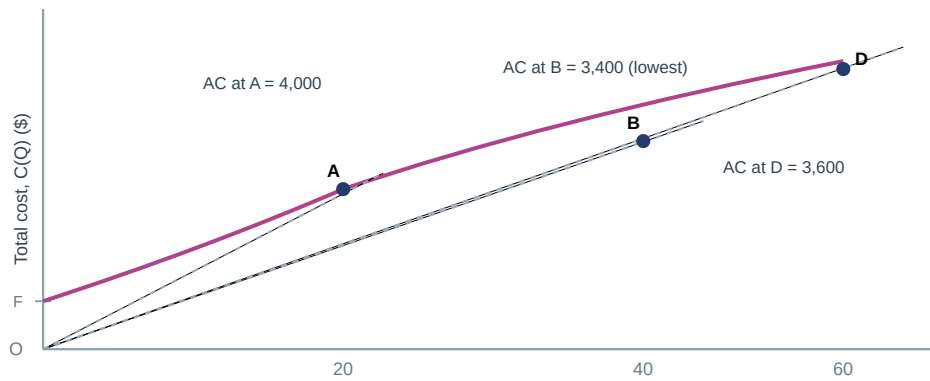
Total cost starts at the **fixed cost F** (the intercept, output = 0) and rises with output. Here it is *convex* (gets steeper) — each extra car costs a bit more than the last, i.e. rising marginal cost.

Average cost, marginal cost, and how they relate

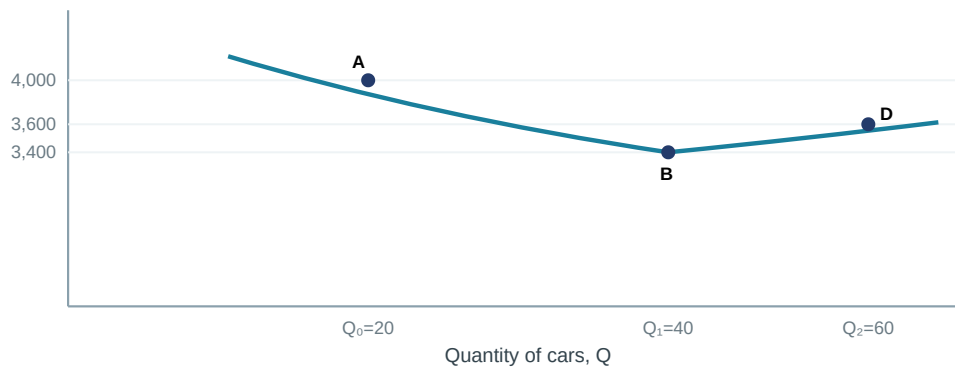
TWO COST MEASURES YOU MUST NOT CONFUSE

- **Average cost (AC)** = cost *per unit* produced = $C(Q)/Q$. Graphically it is the **slope of the ray from the origin** to a point on the total-cost curve.
- **Marginal cost (MC)** = the effect on total cost of producing *one additional* unit. Graphically it is the **slope of the total-cost curve** (its tangent) at a point.

Total cost $C(Q)$: AC = slope of ray from origin O



Average cost of production (\$)



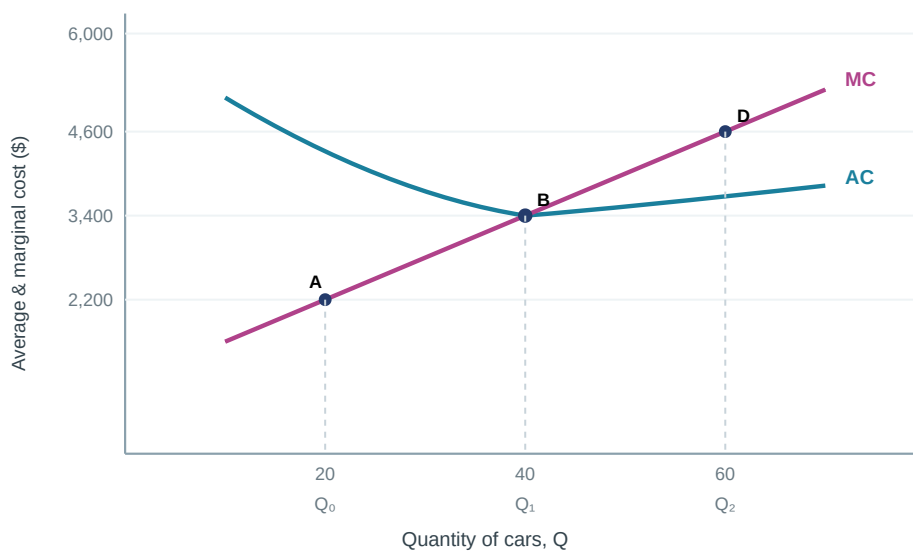
Top: average cost at a point = slope of the ray from the origin O. That slope is smallest at B, so **AC is lowest at B (3,400)**. Bottom: plotting those slopes gives the familiar **U-shaped average-cost curve** — AC falls, then rises (e.g. overtime, machine breakdowns at high output).

THE MC-AC RULE (LEARN THIS COLD)

- If $AC > MC \rightarrow$ average cost is **decreasing** (the cheap extra unit pulls the average down).
- If $AC < MC \rightarrow$ average cost is **increasing** (the expensive extra unit pulls the average up).

Consequence: the MC curve always cuts the AC curve at the AC curve's lowest point.

MC cuts AC at the minimum of AC



To the left of B, MC is below AC $\rightarrow$ AC falling. To the right, MC is above AC $\rightarrow$ AC rising. So MC must pass exactly through B, the **bottom of the AC curve** (here $Q_1 = 40$, cost = 3,400).

Advantages of being large: economies of scale

Many sectors are dominated by a few large firms. A key explanation: **average costs fall as output rises**.

DEFINITIONS — DON'T CONFUSE THESE TWO

- **Economies of scale** — average cost of production *decreases* as output increases. (Large firms get a cost advantage.) Newspaper industry (strong scale economies) vs the restaurant sector (weak).
- **Economies of scope** — cheaper to produce two products *jointly* in one firm than separately in two, usually because a **common input is shared** (e.g. internet + cable TV; insurance + banking).

Economies of scale come from two broad sources — **technological advantages** and **cost advantages**.

1. Technological advantages → returns to scale

Costs are fundamentally linked to the production process. Scaling inputs up can raise output more or less than proportionally:

Returns to scale	Output vs inputs	Effect on average cost
Increasing returns	Output rises <i>more</i> than proportionally to inputs (inputs rise less than proportionally to output)	AC decreasing
Constant returns	Output rises <i>proportionally</i> with inputs	AC constant
Decreasing returns	Output rises <i>less</i> than proportionally to inputs (inputs rise more than proportionally to output)	AC increasing

SOURCES OF INCREASING RETURNS

- Specialization & indivisibilities in the production process
- Engineering reasons — e.g. a gas pipeline's material cost scales with its *circumference*, but capacity with its *cross-sectional area*; container ships, tankers

SOURCES OF DECREASING RETURNS

- Coordination becomes harder at scale
- Supervision / management overheads grow

2. Cost advantages

- **Fixed costs** — when fixed costs are large, spreading them over more units lowers AC as scale rises. (Fixed = doesn't vary with the *production rate*: R&D, IT, advertising. A pharma firm's R&D can vary year to year, but not with how many pills it makes.)
- **Bargaining power** — large firms buy inputs on more favourable terms (e.g. the Ahold–Delhaize merger).

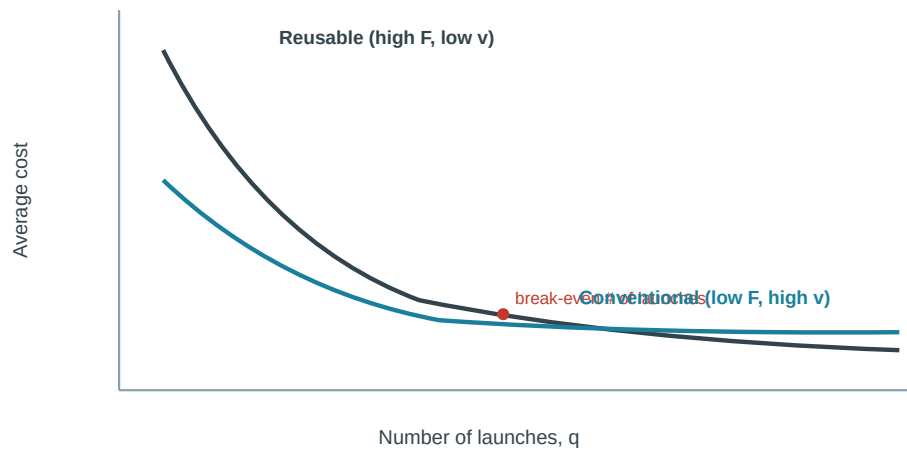
Worked example: how SpaceX makes space travel cheaper

Fixed development cost F and constant variable cost per launch v give a textbook economy of scale:

$$TC(q) = F + v \cdot q \Rightarrow AC(q) = v + \frac{F}{q} \quad (\text{decreases as } q \text{ rises})$$

SpaceX makes rocket boosters **reusable**: this *lowers* variable cost per launch v but *raises* fixed development cost F (engineering the boosters to return). Reusability wins **only if the number of launches is high enough**:

Average cost with and without reusable boosters



At **low** launch counts the reusable design's big fixed cost dominates (higher AC). Past the crossing point, its lower variable cost wins → **reusable is cheaper per launch when volume is high**. Same $v + F/q$ logic as any economy of scale.

C. Demand, elasticity & revenue (bridging — slides not in upload)

WHY THIS SECTION IS HERE

The profit-maximization material in Part 2 constantly uses the **demand curve**, **price elasticity** and **marginal revenue**. These were on the missing middle slides, so here is the standard treatment you need to make Section D make sense.

The demand curve & willingness to pay

DEFINITION — DEMAND CURVE

Shows, for each price, the quantity customers will buy (equivalently, for each quantity, the highest price at which it can all be sold). A firm with **market power** (a differentiated product) faces a **downward-sloping** demand curve: to sell more it must lower the price. Each point on the curve is a customer's **willingness to pay (WTP)** — its reservation price.

Price elasticity of demand

DEFINITION — PRICE ELASTICITY OF DEMAND (ϵ)

The **percentage change in quantity demanded** divided by the **percentage change in price** (taken as a positive number):

$$\epsilon = - \frac{\% \text{ change in quantity}}{\% \text{ change in price}} = - \frac{\Delta Q / Q}{\Delta P / P}$$

- $\epsilon > 1$ → demand is **elastic** (quantity very responsive; cutting price *raises* revenue).
- $\epsilon < 1$ → demand is **inelastic** (quantity barely responds; cutting price *lowers* revenue).
- $\epsilon = 1$ → revenue is at a maximum.

Total & marginal revenue

DEFINITIONS

- **Total revenue** $R = P \cdot Q$.
- **Marginal revenue (MR)** = the change in total revenue from selling *one more* unit = dR/dQ .

WHY MR IS BELOW THE PRICE

To sell one extra unit a firm with market power must **lower the price on all units**. So $MR = (\text{price of the new unit}) - (\text{revenue lost on the units it could have sold dearer})$. That “lost revenue” term makes $MR < P$ whenever demand slopes down. **For a linear (straight-line) demand curve, the MR curve has the same intercept and is exactly twice as steep.**

This last fact is used repeatedly in Section D, so keep it handy: *linear inverse demand* $P = a - bQ \Rightarrow R = aQ - bQ^2 \Rightarrow MR = a - 2bQ$.

D. Pricing & production decisions — profit maximization

Now bring **demand and cost together** to find the profit-maximizing price and quantity. There are **two equivalent methods**, and (obviously) they give the same answer:

1. **Isoprofit curves + demand curve** — the “constrained optimization under scarcity” picture.
2. **Marginal revenue = marginal cost** — the marginal picture.

Method 1 — Isoprofit curves

DEFINITIONS

- **(Economic) profit** = total revenue – total cost: $\pi = PQ - TC(Q) = Q(P - AC)$.
- **Isoprofit curve** — all price–quantity combinations that give the *same* profit (the firm's “indifference curve”).
- The **zero-profit curve is the AC curve** (where $P = AC$, so $\pi = 0$).
- Isoprofit curves **above the AC curve** = positive profits; curves **further from the origin** = higher profit.

SLOPE OF AN ISOPROFIT CURVE

$$\text{slope} = - \frac{P - MC}{Q}$$

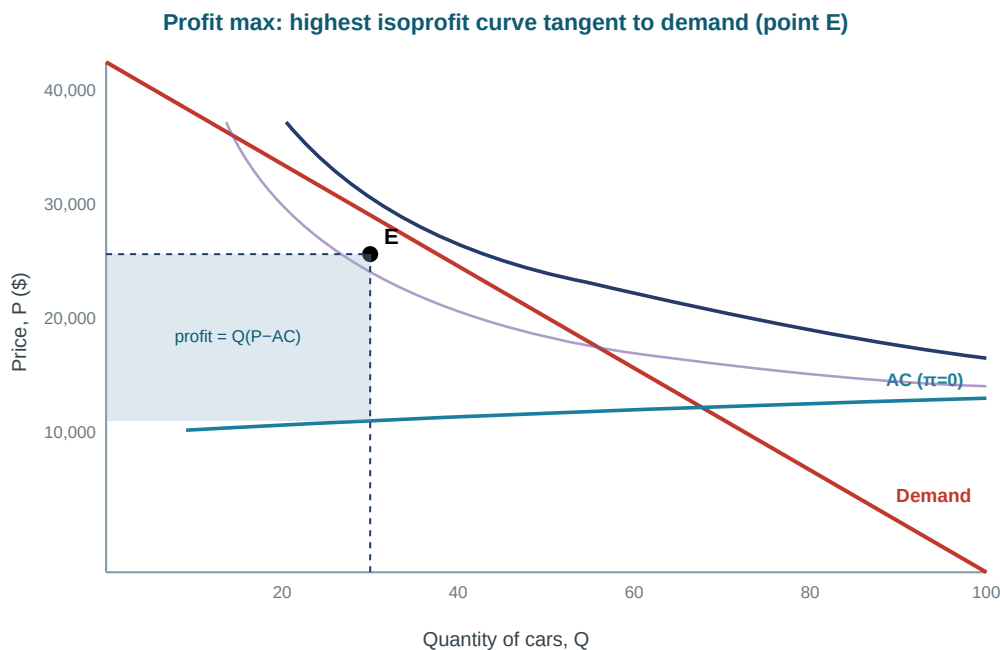
When Q rises by one unit, profit changes by $(P - MC)$; to hold profit constant, price must change by $-(P - MC)/Q$.

- If $MC > P$ (MC above the isoprofit curve) → isoprofit curve slopes **up** (need a higher price to offset the costly extra unit).
- If $MC < P$ (MC below) → isoprofit curve slopes **down**.

Method 1 as constrained optimization (MRS = MRT)

This is exactly the consumer's problem from Unit 3, with new labels:

Consumer (Unit 3)	Firm (Unit 7)
Indifference curves	Isoprofit curves — slope = MRS
Budget constraint / feasible frontier	Demand curve — the feasible frontier; slope = MRT
Optimum where $MRS = MRT$ (tangency)	Profit max where $MRS = MRT$ — highest isoprofit curve <i>tangent</i> to demand



The firm reaches the **highest isoprofit curve that still touches the demand curve** — the tangency **E**. There, the isoprofit slope (*MRS*) equals the demand slope (*MRT*). At **E**, $P > MC$ and $P > AC$, so profit (the shaded rectangle $Q(P-AC)$) is positive.

Profit-maximizing price: the markup rule

Setting $MRS = MRT$ and doing the algebra gives one of the most important results in the course — the **Lerner markup rule**:

$$\frac{P - MC}{P} = \frac{1}{\epsilon}$$

Slope of isoprofit (<i>MRS</i>)	Slope of demand (<i>MRT</i>)	Set $MRS = MRT \Rightarrow$
$-(P - c)/Q$	$-P/(\epsilon Q)$	$(P - c)/Q = P/(\epsilon Q) \Rightarrow (P - c)/P = 1/\epsilon$

READING THE MARKUP RULE (HERE $c = MC$)

The **price markup** (price-cost margin) equals the **inverse of the demand elasticity**. **More elastic demand (larger ϵ) $\Rightarrow$ smaller markup $\Rightarrow$ lower optimal price**. A firm whose customers are very price-sensitive cannot charge much above marginal cost. This single equation links *market power* (Section F) to *elasticity*.

Method 2 — Marginal revenue = marginal cost

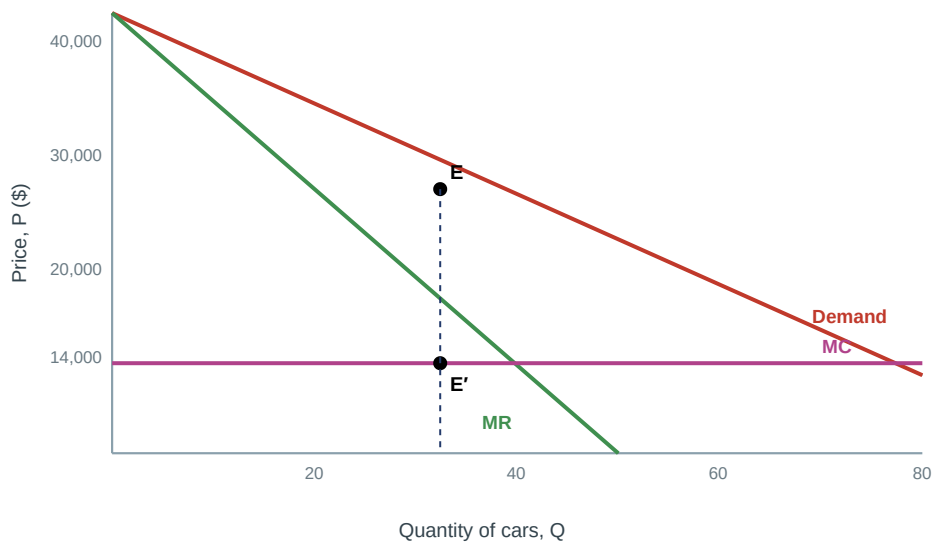
THE $MR = MC$ RULE

To maximize profit, produce at the scale where **marginal revenue = marginal cost**.

- If $MR > MC \rightarrow$ sell more (the extra unit adds more revenue than cost).
- If $MR < MC \rightarrow$ sell less.
- Algebra: $\pi = PQ - TC(Q)$; the first-order condition $d\pi/dQ = 0$ gives $MR = MC$.

Then read the **price off the demand curve** at that quantity. One can show this price also satisfies $(P - MC)/P = 1/\epsilon$ — the two methods agree.

Find Q where $MR = MC$, then go up to demand for P



For a linear demand curve, **MR has the same intercept but is twice as steep**. The profit-maximizing quantity is where **MR meets MC (E')**; the profit-maximizing price is found by going straight up to the demand curve (E). Note E sits above MC — **price exceeds marginal cost**.

Worked numerical example

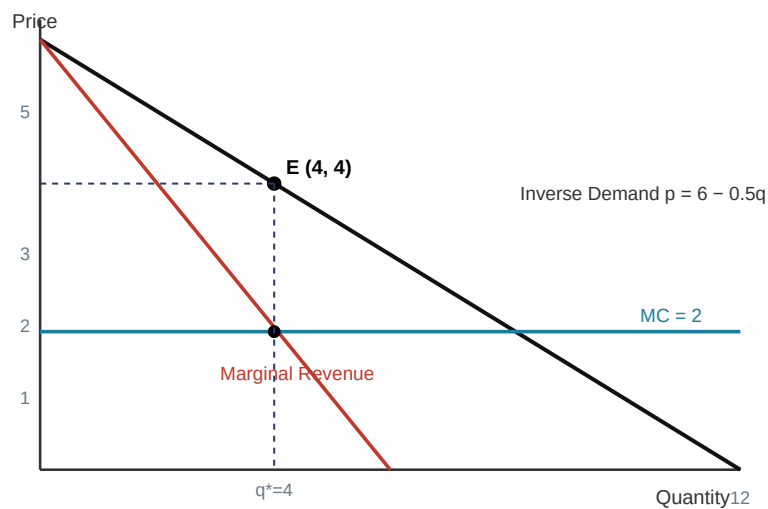
SOLVE IT STEP BY STEP

Demand: $Q = 12 - 2P$, so **inverse demand** $P = 6 - 0.5Q$. **Marginal cost** $MC = 2$.

1. **Total revenue:** $R = PQ = (6 - 0.5Q)Q = 6Q - 0.5Q^2$.
2. **Marginal revenue:** $MR = dR/dQ = 6 - Q$ (straight line, twice as steep as demand — same 6 intercept).
3. **Set $MR = MC$:** $6 - Q = 2 \Rightarrow Q^* = 4$.
4. **Price from demand:** $P^* = 6 - 0.5(4) = 4$.

Answer: $Q^* = 4$, $P^* = 4$. (Check: price 4 > MC 2, as market power implies.)

Example: $p = 6 - 0.5q$, $MR = 6 - q$, $MC = 2$



MR (red) hits MC (teal) at $q^* = 4$; going up to the demand curve gives $p^* = 4$. The gap between price and MC (4 vs 2) is the firm's markup, sustained by its market power.

E. Gains from trade (bridging — slides not in upload)

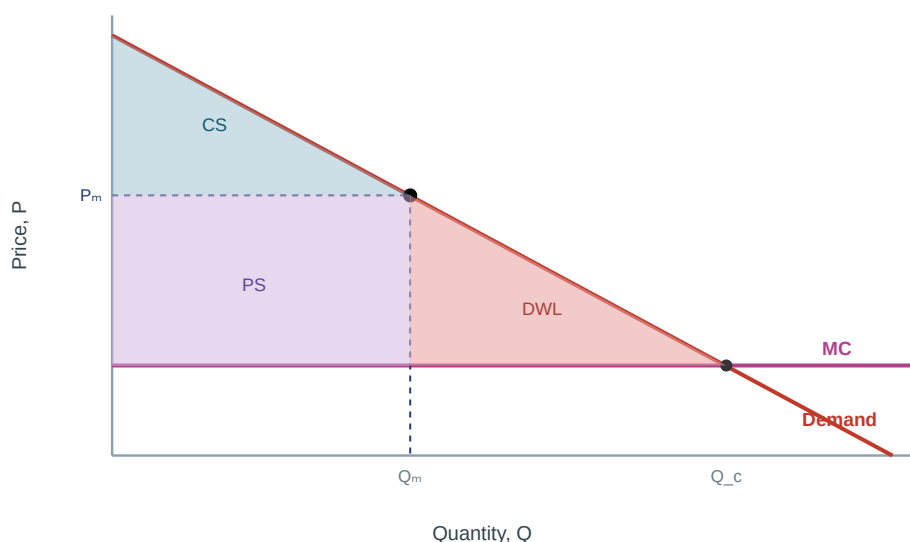
ABOUT THIS SECTION

Part 2 ends on the title slide “Gains from Trade”; its content slides weren't in the upload. Here is the standard CORE/KU Leuven treatment of **surplus** and the **welfare cost of market power**, which is where the unit heads next.

DEFINITIONS — MEASURING THE GAINS FROM TRADE

- **Consumer surplus (CS)** — total *willingness to pay* minus what customers actually pay. Area **below the demand curve**, above the price.
- **Producer surplus (PS)** — revenue minus variable cost of production. Area **above the MC curve**, below the price.
- **Total surplus** = CS + PS — the total gains from trade.
- **Deadweight loss (DWL)** — surplus that is *lost* because trades that would benefit both sides don't happen (e.g. when a firm restricts output to keep price high).

Surplus & the welfare cost of market power



A firm with market power sets price P_m above MC and sells only Q_m (less than the competitive Q_c where price = MC). Consumers keep CS (blue), the firm earns PS (purple). The red triangle is **deadweight loss** — mutually beneficial trades between Q_m and Q_c that never happen. This is the efficiency cost of market power that policy tries to reduce.

F. Competition, differentiation & market power (bridging)

DEFINITION — MARKET POWER

The ability of a firm to **set its price above marginal cost** without losing all its customers. It comes from producing a **differentiated product** (brand, quality, location) so the firm faces a **downward-sloping** demand curve rather than being a price-taker.

HOW IT ALL TIES TOGETHER

The markup rule $(P - MC)/P = 1/\epsilon$ is the bridge:

- **Strong differentiation** → **less elastic demand (small ϵ)** → **large markup, high price** (much market power).
- **More competition / closer substitutes** → **more elastic demand (large ϵ)** → **small markup, price near MC** (little market power).

In the limit of perfect competition, $\epsilon \rightarrow \infty$, the markup $\rightarrow 0$, and price = marginal cost (no deadweight loss).

Policy angle: because market power creates deadweight loss, competition policy, regulation and lowering barriers to entry aim to push ϵ up (more substitutes) and shrink the markup — redistributing surplus toward customers and improving efficiency.

Summary

1. **Costs.** The cost function splits into **fixed** and **variable** costs. $AC = C/Q$ (slope of ray from origin); MC = slope of the cost curve. **MC cuts AC at AC's minimum.**
2. **Economies of scale** (AC falls with output) come from **technological** advantages (increasing returns to scale) and **cost** advantages (spreading fixed costs, bargaining power). Don't confuse with **economies of scope**.
3. **Demand & revenue.** A firm with market power faces a downward-sloping demand curve; $MR < P$, and for linear demand MR is twice as steep.
4. **Profit maximization.** Two equivalent methods: highest **isoprofit curve tangent to demand** ($MRS = MRT$), or $MR = MC$. Both give price via the demand curve, with $P > MC$ and (if profitable) $P > AC$.
5. **Markup rule.** $(P - MC)/P = 1/\epsilon$: markup = inverse elasticity. More elastic demand → lower price.
6. **Gains from trade & market power.** Surplus = CS + PS; market power raises price above MC, creating **deadweight loss**. Competition and policy shrink the markup toward the efficient outcome.

Glossary — quick revision

Term	One-line meaning
Cost function $C(Q)$	How total cost varies with output.
Fixed / variable cost	Doesn't vary with output / increases with output.
Average cost (AC)	Cost per unit, C/Q ; slope of ray from origin.
Marginal cost (MC)	Cost of one more unit; slope of the cost curve.
MC–AC rule	MC below AC → AC falling; MC above → AC rising; they cross at min AC.
Economies of scale	AC decreases as output rises.
Economies of scope	Cheaper to produce two goods jointly (shared input).
Returns to scale	Output response to a proportional rise in all inputs (increasing / constant / decreasing).
Demand curve	Quantity customers buy at each price (downward-sloping under market power).
Price elasticity ϵ	$-(\% \Delta Q)/(\% \Delta P)$; >1 elastic, <1 inelastic.
Marginal revenue (MR)	Revenue from one more unit; $<$ price; twice as steep for linear demand.
Isoprofit curve	Price–quantity combinations giving equal profit; zero-profit curve = AC.
$MRS = MRT$	Profit max: isoprofit slope = demand slope (tangency).
Markup rule (Lerner)	$(P - MC)/P = 1/\epsilon$.
Market power	Ability to price above MC via product differentiation.
Consumer / producer surplus	Gains to buyers / sellers from trade.
Deadweight loss	Surplus lost when beneficial trades don't happen (e.g. under market power).

Self-test questions

1. A firm's AC is 3,600 and its MC is 4,600 at the current output. Is average cost rising or falling? What does this imply about where you are relative to minimum AC?

MC (4,600) > AC (3,600), so AC is *rising* → you are to the *right* of the minimum-AC point. Producing a bit less would lower average cost.

2. Demand is $Q = 20 - 2P$ and $MC = 3$. Find the profit-maximizing quantity and price.

Inverse demand: $P = 10 - 0.5Q$. $R = 10Q - 0.5Q^2 \Rightarrow MR = 10 - Q$. Set $MR = MC$: $10 - Q = 3 \Rightarrow Q^* = 7$. Price: $P^* = 10 - 0.5(7) = 6.5$. (Check $P > MC$.)

3. Two firms have the same MC. Firm A's customers are very price-sensitive ($\epsilon = 4$); Firm B's are loyal ($\epsilon = 1.5$). Which charges the higher markup, and why?

Markup = $1/\epsilon$. Firm A: $1/4 = 25\%$. Firm B: $1/1.5 \approx 67\%$. Firm B charges the higher markup — less elastic demand (more market power) lets it price further above MC.

4. Why does making rocket boosters reusable only lower average cost per launch if the number of launches is large?

Reusability raises fixed development cost F but lowers variable cost v , so $AC = v + F/q$. At small q the larger F/q dominates and AC is higher; only once q is large enough does the lower v outweigh the spread-out fixed cost, so the reusable AC curve dips below the conventional one.

5. At the profit-maximizing point, is price above or below marginal cost? Above or below average cost? What does each tell you?

Price > MC always for a firm with market power (that gap is the markup). Price vs AC determines profitability: $P > AC$ means positive economic profit; $P = AC$ means zero profit; $P < AC$ means a loss.